

Testing Replication With Open-Source Data

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Abstract

This project leveraged open-source data and open-source-software for replication research in psychological science. Findings included:

- Confirmatory Factor Analysis of the 21-item Depression Anxiety Stress Scale (DASS-21)
- Gender differences in personality traits
- Gender differences in experiencing depression, anxiety, and stress
- Personality as a predictor of depression, anxiety, and stress for males and females



Introduction & Motivation

Problem & question: In 2011, high rates of replication failure plunged psychological research into a crisis of public trust. This project aimed to repair that trust by testing replication using open-source data and open-source software.

Impact & stakeholders: Replicating findings of gender differences in personality and depression, anxiety, and stress will improve the knowledge base of clinicians who provide mental health treatment.



Background & Related Work

- Replication research is vital for improving our understanding of human behavior (Youyou et al., 2023).
- Replication with slight methodological changes yields equally rigorous results, and advances the psychological sciences (Nosek & Errington, 2020).
- Target studies for replication found:
 - Females tend to rate lower in emotional stability and higher in agreeableness compared to males (Weisberg et al., 2011);
 - Emotional stability is the best predictor of depression, stress, and anxiety (Uliaszek et al., 2010);
 - Confirmatory Factor Analysis (CFA) suggests DASS-21 is suitable for comparison of means between males and females (Gomez et al., 2014)



Methods & Evaluation

Approach: Analyses were conducted in R using R Studio, included R's built in independent samples t-test, and packages such as lavaan, and effectsize

Validation:

- Participants who failed an attention check were removed
- Given the large sample size:
 - Independent sample t-tests were reported by effect size
 - CFA was reported in terms of Comparative Fit Index (Furr, 2018)



Reproducibility & Sources

Repository: <https://ballen-git.github.io/capstone.html>



Results & Visualizations

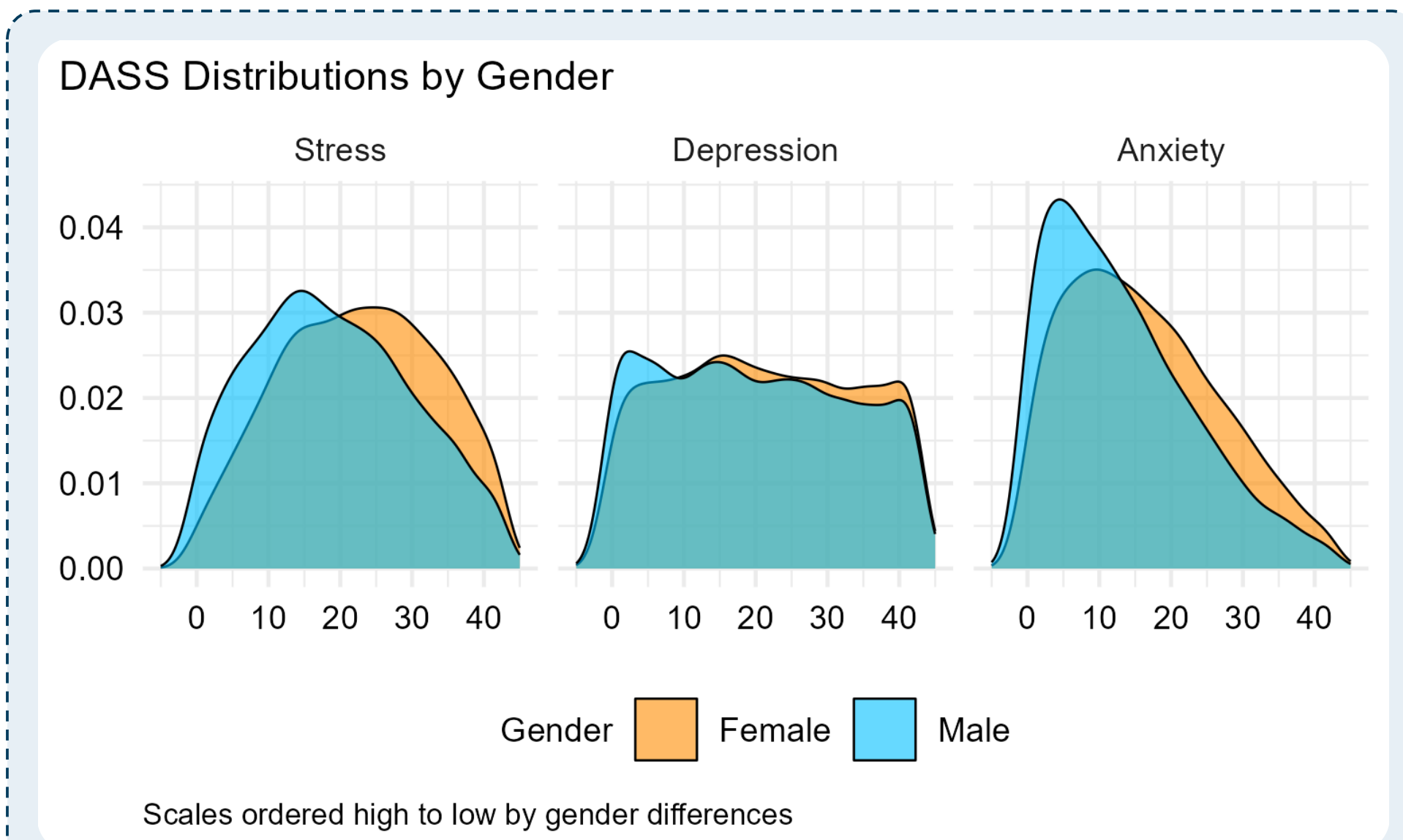


Fig 1. Distribution of DASS-21 scores by gender. Female scores trend higher than male scores, especially for stress and anxiety.

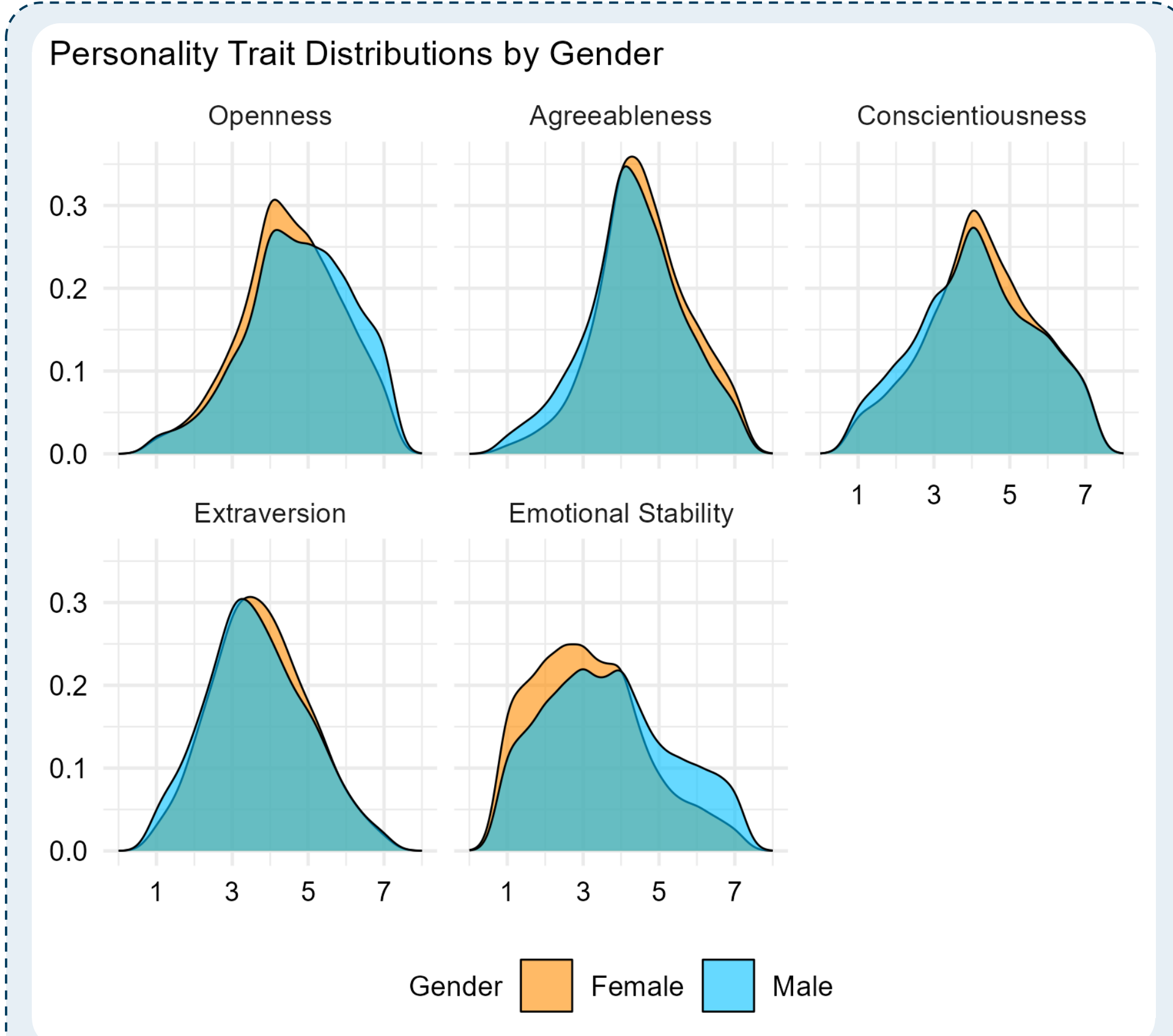


Fig 2. Distribution of personality scores by gender with small to moderate differences. Females rate higher on trait agreeableness and lower in emotional stability than males.

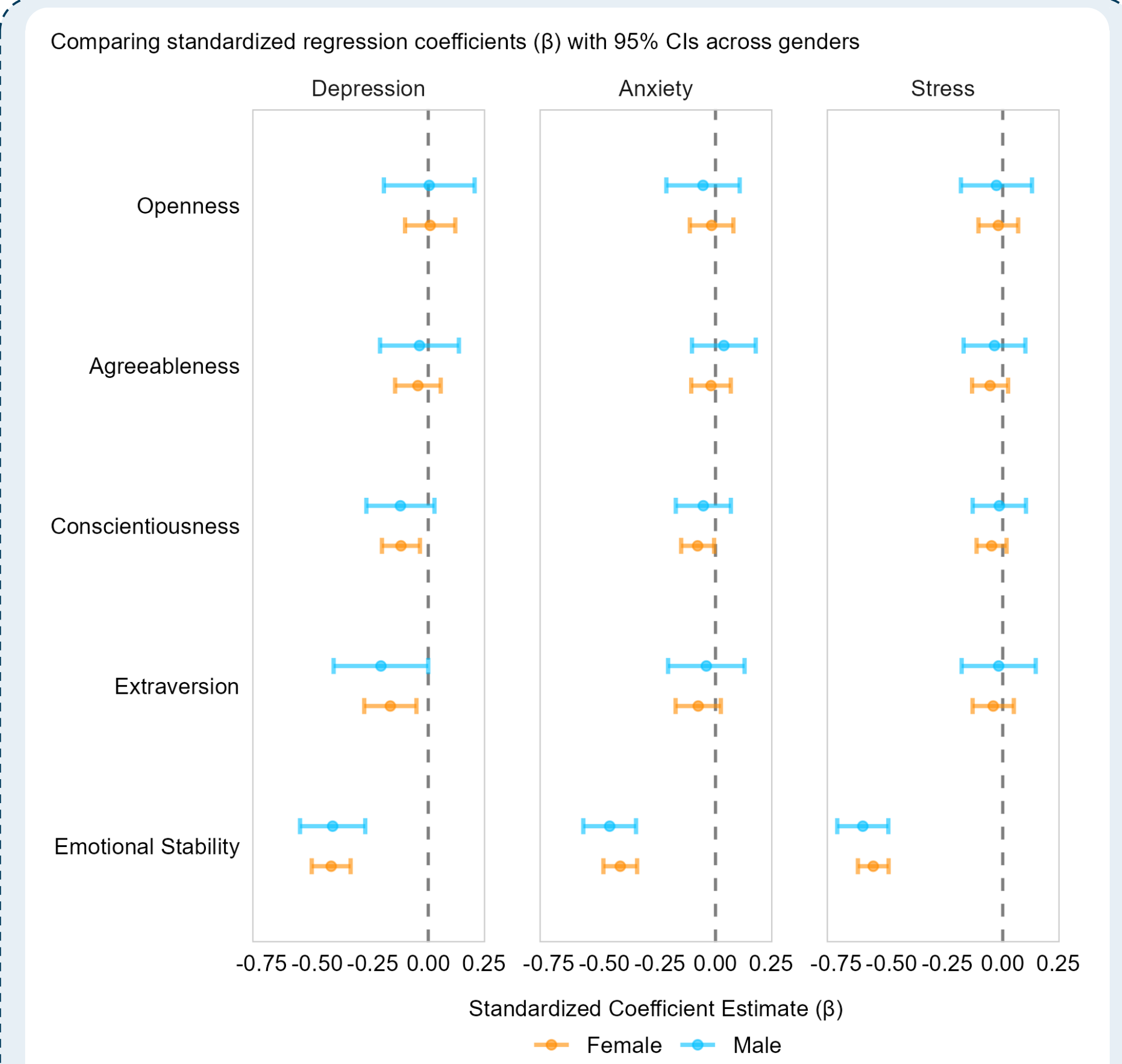


Fig 3. Emotional stability is the most influential trait for personality as a predictor of stress, depression, and anxiety for both males and females.



Results & Visualizations

- Females rated higher on trait agreeableness and lower on emotional stability compares to males ($d = .19$, $d = -.35$).
- Compared to males, females reported higher levels of depression, anxiety, and stress ($d = .10$, $d = .31$, $d = .34$, respectively).
- Emotional stability was the most influential trait for negative emotions.

Table 8

Gender Differences (d) in Big Five Personality Traits Study Comparisons

Mental Health Indicator	Studies		Studies		Present Study			
	Sambol et al. (2022)	R^2	β	R^2	β	R^2	β	R^2
Personality Dimension						Female	Male	
Depression	.36			.26		.31	.34	
Openness to Experience		.08			0	.01		0
Conscientiousness		.18			.17	.12		.13
Extraversion		.09			.17	.17		.21
Agreeableness		.10			.14	.05		.04
Emotional Stability		.49			.34	.44		.43
Anxiety	.28			.27		.24	.27	
Openness to Experience		.01			0	.02		.06
Conscientiousness		.11			.12	.08		.05
Extraversion		.04			.07	.08		.04
Agreeableness		.08			.06	.02		.04
Emotional Stability		.48			.45	.43		.48
Stress	.42			.37		.40	.43	
Openness to Experience		0			.05	.02		.03
Conscientiousness		.10			.05	.05		.03
Extraversion		.07			.03	.04		.02
Agreeableness		.06			.08	.06		.04
Emotional Stability		.62			.57	.58		.63

Note: Where emotional stability was reported as neuroticism, the β values have been sign changed. Bolded values indicate statistical significance.

Table 7

Gender Differences (Cohen's d) in Big Five Personality Traits Study Comparisons

Personality Dimension	Studies		
	Lippa et al. (2010)	Weisberg et al. (2011)	Present Study
Openness to Experience	.01	.27	.16
Conscientiousness	.09	.06	.09
Extraversion	.11	.08	.07
Agreeableness	.40	.48	.19
Emotional Stability	-.34	-.39	-.35

Note: Positive effect sizes occur when women score higher than men. Negative effect sizes occur when men score higher than women. Bolded d values indicate statistically significant effect sizes. Where emotional stability was reported as neuroticism, Cohen's d values have been sign changed.



Discussion, Limitations & Conclusions

Discussion: Results supported prior findings, adding to the "credibility revolution" described by Korbacher et al. (2023). Therefore, this study adds to the growing body of replication studies advancing the cause of repairing public trust in the psychological sciences.

Limitations & future work: This survey was conducted online and may have been completed multiple times by the same participant. Additionally, populations with limited or no access to the internet were under sampled. Results from this study will be used to isolate important features from machine models that predict depression, stress, and anxiety in males and females.

Conclusions: This study showed open-source software can be used to conduct rigorous analysis of open-source data to replicate findings from previous research, thus opening the field of psychological science to a broad swath a researchers looking to contribute to the credibility revolution.



Data, Engineering & Ethics

Sources & scale: Open-Source Psychometric Project. The data set contained 39,775 responses to the DASS-21 (Lovibond & Lovibond, 1995) and the Ten-Item Personality Inventory (Gosling et al., 2003).

Pipeline: CSV → R → R Markdown HTML → PDF → Git Hub

Ethics & responsible use: Data was collected and cleaned by Open-Source Psychometric Project for public use. There are no known ethical concerns.

